

Predictive Model Plan – Geldium Delinquency Prediction

1. Model Logic

The goal of the predictive model is to forecast whether a customer is likely to become delinquent. Since the target variable, **Delinquent_Account**, has two possible outcomes (0 = Non-Delinquent and 1 = Delinquent), Logistic Regression is selected as the primary modeling approach.

Top 5 Input Features

- **Credit_Utilization:** Higher credit utilization may indicate greater use of available credit and increased financial pressure.
- **Debt_to_Income_Ratio:** A higher ratio may indicate a greater debt burden relative to income.
- **Credit_Score:** Provides information about the customer's overall credit profile.
- **Missed_Payments:** Provides information about the customer's previous repayment behavior.
- **Month_4 Payment Status:** Recent payment behavior, particularly late payments, may provide an early warning signal.

Conceptual Model Workflow

Customer Data → Feature Selection → Data Preparation → Logistic Regression → Delinquency Probability → Risk Classification → Prediction Output

The model would process relevant customer financial and payment information and estimate the probability that a customer will become delinquent. A selected probability threshold could then be used to classify customers as predicted delinquent or non-delinquent.

Step-by-Step Process

1. **Data Input:** Customer financial, credit, employment, and payment information is provided as input.
2. **Feature Selection:** Relevant features such as credit utilization, debt-to-income ratio, credit score, missed payments, and recent payment status are selected based on the EDA findings.
3. **Data Preparation:** The variables are prepared in a format suitable for the model, using the cleaned dataset developed during the EDA stage.
4. **Model Training:** A Logistic Regression model would be trained using historical customer records and their known delinquency outcomes.
5. **Probability Prediction:** The model would generate a probability representing the likelihood of customer delinquency.
6. **Risk Classification:** A suitable probability threshold would be applied to classify customers into predicted delinquent or non-delinquent groups.
7. **Output:** The resulting risk predictions could help Geldium identify customers who may require additional monitoring or review.

2. Justification for Model Choice

Logistic Regression is selected because the objective is to predict a binary outcome: whether a customer is delinquent or non-delinquent. The model is relatively easy to interpret, computationally efficient, and straightforward to implement and monitor. Its coefficients can provide an indication of how the input variables are associated with the predicted delinquency probability, which is valuable in financial services where transparency and explainability are important.

Logistic Regression also provides a strong and understandable baseline before considering more complex models such as decision trees or ensemble methods. However, one limitation is that it may not capture complex non-linear relationships as effectively as more advanced models. Therefore, other models could be considered in a later stage if they provide meaningful performance improvements while maintaining appropriate explainability.

3. Evaluation Strategy

The model should be evaluated using multiple performance metrics rather than relying only on accuracy because the dataset contains 80 delinquent customers (16%) and 420 non-delinquent customers (84%).

Evaluation Metrics

- **Accuracy:** Measures the overall percentage of correct predictions. However, accuracy should not be considered alone because the target classes are imbalanced.
- **Precision:** Measures how many customers predicted as delinquent are actually delinquent. Higher precision means fewer non-delinquent customers are incorrectly identified as delinquent.
- **Recall:** Measures how many actual delinquent customers are correctly identified. Recall is important because failing to identify a potentially delinquent customer may create greater financial risk.
- **F1 Score:** Combines precision and recall into a single measure and is useful when the target classes are imbalanced.
- **ROC-AUC:** Measures how well the model distinguishes between delinquent and non-delinquent customers across different classification thresholds.
- **Confusion Matrix:** Shows true positives, true negatives, false positives, and false negatives. It helps understand the types of errors made by the model.

Fairness and Bias Checks

The model should also be evaluated for potential bias. Model performance and error rates can be compared across relevant customer groups, such as employment status or location, to identify whether certain groups experience substantially different outcomes.

Potential checks include:

- Compare precision and recall across customer groups.
- Compare false-positive and false-negative rates.
- Check whether the model disproportionately identifies a particular group as high risk.
- Review whether any features could introduce inappropriate or unintended bias.
- Monitor model performance and fairness over time.

Ethical Considerations

Because the model would be used to assess customer financial behavior, predictions should be treated as decision-support information rather than an automatic final decision. Geldium should ensure that predictions are explainable, appropriately monitored, and reviewed for potential bias. Sensitive or inappropriate customer characteristics should not be used as predictive features without a valid business and regulatory justification.

GenAI Usage

Generative AI was used to help structure the predictive modeling approach, compare possible model types, identify suitable evaluation metrics, and refine the fairness and ethical considerations.

Example GenAI Prompts Used:

- *"Outline a predictive modeling pipeline to forecast credit delinquency, from feature selection to model evaluation."*
- *"Suggest two modeling options for predicting delinquency and explain the trade-offs between them."*
- *"Explain the advantages and disadvantages of Logistic Regression for financial risk prediction."*
- *"Suggest metrics to evaluate a financial risk prediction model for accuracy, fairness, and bias."*
- *"Explain how to check for bias and unfair treatment across customer groups in a credit risk model."*

Conclusion

For the Geldium delinquency prediction problem, Logistic Regression provides an interpretable and practical conceptual approach for estimating delinquency risk. The proposed model would use relevant financial and payment features identified during EDA and generate a probability-based delinquency prediction. Model performance would be evaluated using accuracy, precision, recall, F1 score, ROC-AUC, and a confusion matrix, along with fairness and bias checks. More complex models could be considered later if they provide meaningful performance improvements while maintaining transparency and responsible use of financial predictions.